

Economics

Demand

Demand = Purchasing Power + Desire/Wish/Will

* Law of Demand:-

Other things remaining the same when the price of something increases its demand decreases and when the price decreases its demand increases.

* Symbols:-

Variables	Symbols	Variables	Symbols
Cost	C	Quantity	Q
Price	P	Export	X
Saving	S	Import	M
Utility	U	Income	Y
		Profit	$\pi (P_i)$

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★ Demand Schedule:-

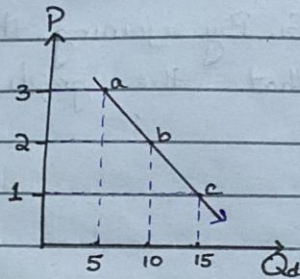
Price	Demand (KG)
1 Rs	15
2 Rs	10
3 Rs	5

Explain:-

The above demand schedule shows the demand of consumer at different price levels. When the price is 1Rs per KG the demand is 15KG. As the price increases to 2Rs per KG the demand decreases to 10KG. When the price increases to 3Rs per KG the demand decreases to 5KG.

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* Demand Graph:-



Point	Price	Demand
Point a	= 3Rs	, 5KG
Point b	= 2Rs	, 10KG
Point c	= 1Rs	, 15KG

Explain:-

In this diagram. We measure price along y-axis and demand along x-axis. From this diagram we can see that at point 'a' when the price is 3Rs per KG the demand is 5KG but at point 'b' when the price decreases to 2Rs per KG the demand increases to 10KG.

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At point 'c' when the price decreases to 1Rs per KG the demand increases to 15 KG. By joining the point a, b and c we can see that the graph is negatively sloped.

*Market Demand:-

In the market there are many consumers demanding the commodity at different price levels. When we sum the quantity demanded of the commodity at different prices of all the consumers in the market, we arrive at the "market demand" of the commodity.

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* Market Demand Schedule:-

Price (Per KG)	A	B	C	Market Demand
3	2	3	5	10
2	4	5	11	20
1	6	10	14	30

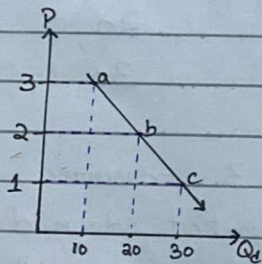
Explain:-

It is assumed that there are 3 consumers in the market, A, B and C. The market demand changes as the price changes. As the price is 3Rs per KG the market demand is 10KG. When the price decreases to 2Rs per KG the market demand increases to 20KG as the consumers A, B and C increase

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their purchase. When the price decreases to 1Rs per KG the market demand increases to 30KG.

★ Market Demand Graph:-



Point	Price	Market D.
Point A	3Rs	10KG
Point B	2Rs	20KG
Point C	1Rs	30KG

Explain:-

In this diagram. We measure price along y-axis and marked demand/quantity demand on x-axis. On point 'a'

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when the price is 3Rs per KG the demand is 10KG. On point 'b' when price decreases to 2Rs per KG the market demand increases to 20KG. On point 'c' when the price decreases to 1Rs per KG the market demand increases to 30KG. When we join the point a, b and c ~~they~~ it will make a slope from upward to downward which is negatively sloped.

Main Characteristics of The Demand Curve

1) It shows the inverse relationship between price and quantity demand. $Q_d = f(p)$. Here price is independent while quantity demand is dependent. It depends upon price. ($\uparrow P = Q_d \downarrow$ or $\downarrow P = Q_d \uparrow$).

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* Why Demand Curve is Negatively Sloped:-

- (i) Entry of new buyers (price effect)
- (ii) Income effect
- (iii) Substitution effect
- (iv) Diminishing marginal utility

Entry of New Buyers:-

Fall in the price of product attracts the new buyers in the market. The people hesitating to purchase the product at higher price, now start to buying it. In the result the demand of that product increases.

Income Effect:-

When the price of product decreases,

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purchasing power of the consumer increases. Thus more units are demanded at low price with same amount of money.

Substitution Effect:-

When the price of a product decreases while the price of its substitute remain the same, the consumer feel that now this product is relatively cheaper, its demand increases while the demand of its ~~sub~~ substitute decreases.

Diminishing marginal ~~effect~~ Utility:-

According to "Law of diminishing marginal utility" every new unit gives less utility than the previous. So additional units can be demanded only at lower price.



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Exceptions & Limitations

Under certain circumstances, the price and demand of the product move in the same direction.

- (1) Giffon Goods (2) Scarce Goods (3) Emergency
(4) The Most Precious Goods (5) Ignorance & Carelessness

* Giffon Goods:-

A giffon good is a low income, non-luxury product that defies standard economic and consumer demand theory. Demand for giffon goods rise when the price rise and falls when the price falls.

* Scarce Goods:-

When people expect that in future there



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will be shortage in the supply of product and it will not be available in the market. They may purchase it's more units even though price is rising.

★ Emergency:-

Emergency goods are those products which a consumer buys in case of an emergency or an absolute urgent requirement.

★ The Most Precious Goods:-

The most precious goods i.e. Diamonds, Stones, Expensive Cars etc which are used as a symbol of status. The demand of such products/goods increase with the increase in price.

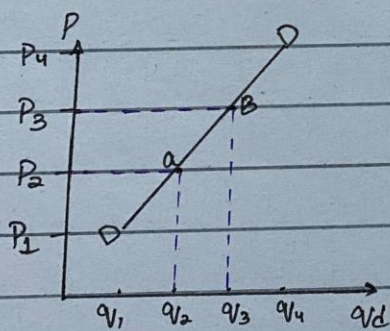


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* Ignorance & Carelessness:-

Some people are ignorant with respect to the price and supply of goods in the market. So, they may purchase more units of goods at high price and less unit at low price.

Exceptional Demand Curve



As the price rise from P_2 to P_3 the quantity demand also increases from q_2 to q_3 . Thus DD "exceptional curve" is positively sloped.

Functional Equation of Demand

* In view of "law of demand" the functional relationship between price and quantity demanded may be described as under.

$$\rightarrow Y = P_1 q_1 + P_2 q_2 + P_3 q_3 + \dots + P_n q_n$$

- i) 'Y' = the amount consumer wants to spend (total money)
- ii) $P_1, P_2, P_3 \dots P_n$ are the price of different goods.
- iii) $q_1, q_2, q_3 \dots q_n$ represent the quantity demanded of these goods.

* According to law of demand.



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- i) P = price of the product
- ii) N = number of consumers
- iii) Y = Income of consumers
- iv) T = Taste, fashion, habit etc of the consumer
- v) π = Price of substitutes

Thus.

$$q_d = f(P, N, Y, T, \pi, \dots)$$

(Quantity demand is function of Price, Number of Consumers, Income, Taste and Price of substitutes)



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Graphic Representation of Demand Function:-

* When

$$q_d = 10 - 2p$$

Here the p (price) is an independent variable.

It gives the value such as 1, 2, 3. To find the value of q we put the values of p in above equation.

$$q = 10 - 2p$$

- i) $q = 10 - 2(1) \rightarrow 10 - 2 = 8$
- ii) $q = 10 - 2(2) \rightarrow 10 - 4 = 6$
- iii) $q = 10 - 2(3) \rightarrow 10 - 6 = 4$



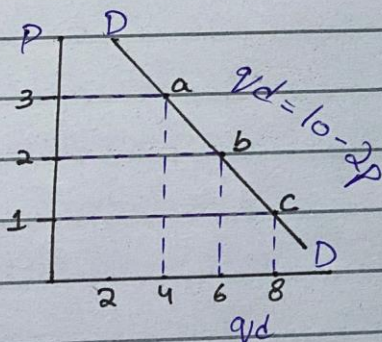
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★ Schedule:-

p	1	2	3
q/d	8	6	4

★ Graph



P is measured along y-axis and q/d is measured along x-axis. There are three points a, b and c. By joining all these points together we derive DD demand curve which is negatively sloped.

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Derivation of Demand Equation

* If demand schedule is given we can derive the demand equation with the help of following Formula:-

$$q - q_1 = \frac{q_2 - q_1}{P_2 - P_1} (p - p_1)$$

- i) q = symbol of quantity demand
- ii) q_1 = Initial quantity demand
- iii) q_2 = changed quantity demand (2nd one)
- iv) p = symbol of price
- v) p_1 = Initial price
- vi) p_2 = Changed (2nd) price

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* According to given schedule

P	$\overset{P_1}{5}$	$\overset{P_2}{10}$	15
qd	$\underset{q_1}{40}$	$\underset{q_2}{30}$	20

Putting in formula:-

$$(q - 40) = \frac{30 - 40}{10 - 5} (p - 5)$$

$$q - 40 = -\frac{10}{5} (p - 5)$$

$$q - 40 = -2(p - 5)$$

$$q - 40 = -2p + 10$$

$$q = -2p + 10 + 40$$

$$q = -2p + 50$$

$$q = 50 - 2p$$

This is the required demand equation.

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Shift in Demand Curve

★ Demand for a product depends upon many factors such as:-

- i) Price of the product.
- ii) Income, Taste, Fashion, Habits, Traditions and Number of consumers.
- iii) Price of substitutes and complementary goods etc.

★ A change in any of the above factor can cause the change in demand.

- a) Expansion or Contraction of demand curve
- b) Rise and Fall in demand OR Shift in demand curve



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* Expansion:-

When the demand increases due to the decrease in price. This is called "expansion".

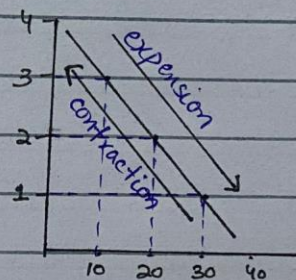
* Contraction:-

When the demand decreases due to the increase in price. This is called "contraction".

* Schedule

P	Qd
3	10
2	20
1	30

* Graph



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★ Rise in Demand:-

When the price of a product increases not due to change in price but due to change in other factors such as: Number of consumers, Income, Price of substitute, Taste, Habit etc. This is called rise in demand. Under these conditions new demand curve will shift upward.

Conditions:-

This upward shift in demand curve may be shown under two conditions.

- i) Rise in qd at the same price.
- ii) Rise in price at the same qd.

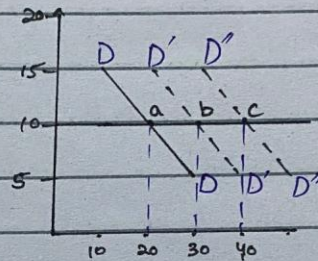


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Rise in Demand

* More qd at same price:-

P	qd
10	20
10	30
10	40



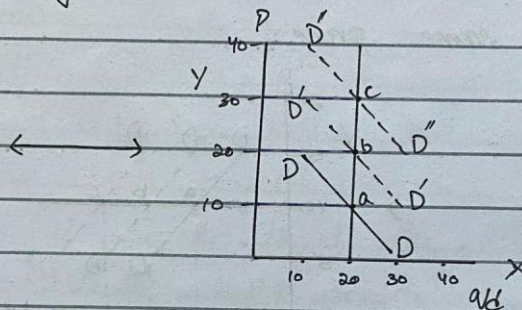
Explain:-

As we can see in the graph, when the price is 10 while qd is 20 the DD cross the point a. When the price remain the same while qd increase to 30 the $D'D'$ line shift to upward right crossing the point b. And same for the $D''D''$, the price 10, qd 40, $D''D''$ cross point c and move upward right.

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* Same qd at higher price:-

P	qd
10	20
20	20
30	20



Explain:-

DD is original demand curve. When the price increases from 10 to 20 but the qd remained unchanged the $D'D'$ moved upward to point b. And again when the price increased but qd remained same a new $D''D''$ formed because DD moved upward to point c.

* The shift of DD in above mentioned "Two Ways" indicate the "Rise in Demand"

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Fall in Demand:-

When demand for a product decrease at the same price or fall in price show no effect on qd (due to the pressure of other factors assumed to remain the same), is known as "Fall in Demand".

Under these conditions new demand curve ~~will~~ will shift downward as $D'D'$ and further as $D''D''$.

Conditions:-

There are two conditions.

- i) Decrease in quantity demand at the same price.
- ii) Quantity demand remains the same at lower price.

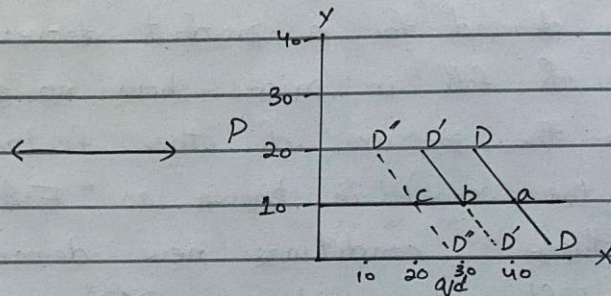


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Fall in Demand

* Less q/d at same price:-

P	q/d
10	40
10	30
10	20



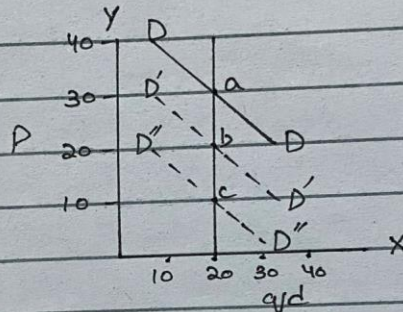
Explain:-

DD is the original demand curve at point a. When the q/d decreases while the price remain the same, the DD shift to left forming new curve D'D'. And same when q/d decreases to 20 while price remain the same, the D'D' move downwardly left forming new curve D''D''.

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* Same qd at lower price:-

P	qd
30	20
20	20
10	20



Explain:-

DD is the original demand curve at point a. When price decreases from 30 to 20 but qd remained same, the DD curve move downwardly forming new curve D'D'. And again when price decreases to 10 from 20, the D'D' curve shift downwardly to point 'c' forming D''D''.

* These "Two ways" shows/indicate the "Fall in Demand".

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Elasticity of Demand

- ★ According to "Law of Demand" it is clear clear that a change in price result an inverse change in quantity demanded. An increase in price shows "contraction in demand" while decrease in price shows the "expension in demand".
- ★ The concept of elasticity was initially introduced by "Alfred Marshall" in his well known book "Principles of Economics."
- ★ Elasticity of demand is the rate of change in quantity demanded in response to change in price.



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★ According to Stonier and Hague "Elasticity of demand is a technical term used by economists to describe the degree of responsiveness of the demand for a good to the change in its price".

★ The demand for all goods is not equally elastic. It may be under:-

- 1) More elastic demand
- 2) Less elastic demand
- 3) Zero elasticity^{of} demand (perfectly in-elastic)
- 4) Infinite elasticity (perfectly elastic)



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*** More Elastic Demand:-**

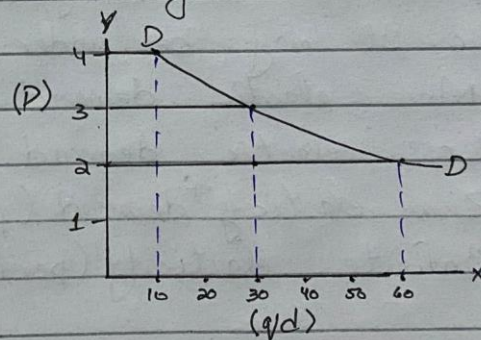
A relatively smaller change in price resulting a big change in quantity demand for a good is known as "more elastic demand good".

More elastic demand curve falls slowly and becomes more flatter.

Schedule

P	q/d
4	10
3	30
2	60

Diagram



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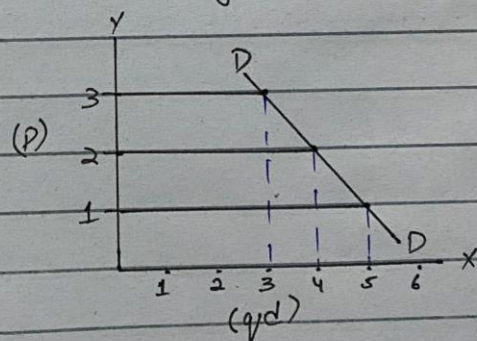
* Less Elastic Demand:-

A relatively large change in price resulting in smaller change in quantity demanded for a good is known as "less elastic demand good". Less elastic demand curve falls sharply downwardly and become steeper.

Schedule

P	q/d
3	3
2	4
1	5

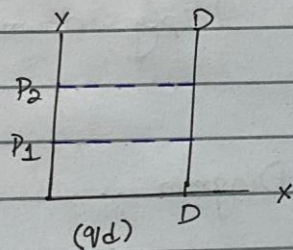
Diagram



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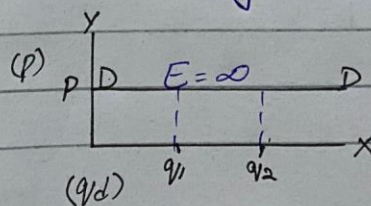
★ Zero Elasticity of Demand:-

When demand does not change even after a big change in price. The demand curve become vertical toward ~~not~~ x-axis.



★ Infinite Elasticity:-

When demand goes on increasing at the same price. Demand curve become perfectly elastic and parallel to x-axis.



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Measurement of Elasticity of Demand

There are three methods to measure the elasticity of demand.

- 1) Unitary Method OR Total Outlay Method
- 2) Flux Method OR Percentage Method
- 3) Formula Method

★ Unitary Method OR Total Outlay Method:-
According to this formula/method

$$E_d = \frac{\text{Proportional Change in Quantity Demanded}}{\text{Proportional Change in Price}}$$



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It may take three forms:-

- a) $E = 1$ b) $E > 1$ c) $E < 1$

a) $E = 1$

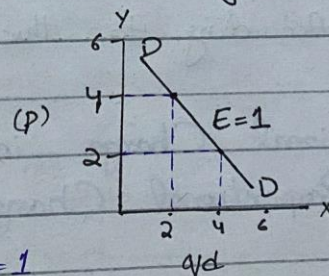
When price increase or decrease but the TR remain same then elasticity of demand is unity. ($E = 1$)

Schedule

P	Qd	TR
4	2	8
2	4	8

$\downarrow P$ or $P \uparrow \rightarrow TR(\text{same}) \rightarrow E_d = 1$

Diagram



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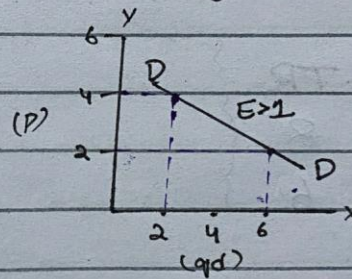
b) $E > 1$

When price decreases then TR increases or when price increases then TR decreases then elasticity of demand is greater than unity ($E > 1$)

Schedule

P	Q	TR
4	2	8
2	6	12

Diagram



$\left. \begin{array}{l} \downarrow P \rightarrow \uparrow TR \\ \uparrow P \rightarrow \downarrow TR \end{array} \right\} \rightarrow E > 1$

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c) $E < 1$

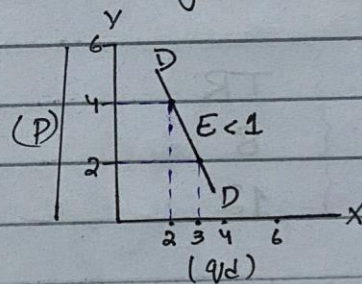
When price decrease then TR also decrease or when price increase then TR also increase then elasticity of demand is less than unity ($E < 1$).

Schedule

P	Q	TR
4	2	8
2	3	6

$\downarrow P \rightarrow \downarrow TR$
 $\uparrow P \rightarrow \uparrow TR$ } $E < 1$

Diagram



2) Flux Method OR Percentage Method

This method has been introduced by Flux.

According to it :-

$$E_d = \frac{\text{Percentage Change in Quantity Demanded}}{\text{Percentage Change in Price}}$$

a) $E = 1$

When the % change in quantity demanded is equal to % change in price then elasticity of demand is unity.

i.e.

$$P = 20\% \text{ change}, E_d = \frac{20}{20} = 1$$

$$Q = 20\% \text{ change}$$



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b) $E > 1$

When % change in quantity demanded is greater than the % change in price then elasticity of demand is greater than the unity.

i.e.

$$P = 20 \% \text{ change} \quad , \quad E_d = \frac{40}{20} = 2 > 1$$

$$Q = 40 \% \text{ change} \quad 20$$

c) $E < 1$

When the % change in quantity demanded is less than the % change in price then elasticity of demand is less than the unity.

i.e.

$$P = 20 \% \text{ change} \quad , \quad E_d = \frac{10}{20} = 0.5 < 1$$

$$Q = 10 \% \text{ change} \quad 20$$



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3) Formula Method

Elasticity of demand may be measured according to the following formulas:-

- i) Arc elasticity formula
- ii) Point elasticity formula

i) Arc elasticity formula:-

When there is a major change in price and quantity demanded then we use arc elasticity formula.

* Formula:-

$$E_d = \frac{\frac{q_0 - q_1}{q_0 + q_1}}{\frac{p_0 - p_1}{p_0 + p_1}} \quad \text{or} \quad E_d = \frac{q_0 - q_1}{q_0 + q_1} \times \frac{p_0 + p_1}{p_0 - p_1}$$

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ii) Point Elasticity Formula:-

When there are small changes in quantity demanded and price then we use point elasticity formula.

★ Formula:-

$$E_d = \frac{\frac{\Delta q}{q}}{\frac{\Delta P}{P}} \quad \text{OR} \quad E_d = \frac{\Delta q}{\Delta P} \times \frac{P}{q}$$

$$\Delta q = q_1 - q_0$$

$$\Delta P = P_1 - P_0$$

$$q_1 = 2^{\text{nd}} Q_d, \quad q_0 = 1^{\text{st}} Q_d$$

$$P_1 = 2^{\text{nd}} \text{ Price}, \quad P_0 = 1^{\text{st}} \text{ Price}$$

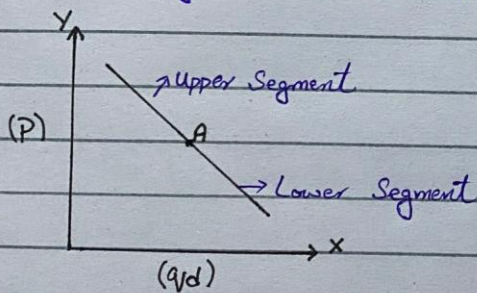
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Geometric Method

In this method we use the following formula:-

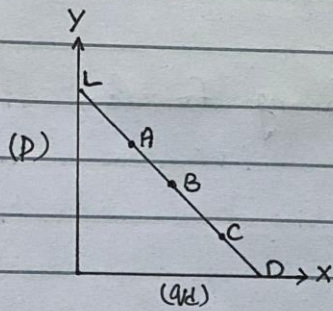
$$E_d = \frac{\text{Length of Lower Segment}}{\text{Length of Upper Segment}}$$



The left side area from point A is called upper segment and the right side area from point A is called lower segment.

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i) At point L :-

$$E_d = \frac{LD}{0} = \infty, \text{ As according to diagram, the lower}$$

segment length is equal to LD while the length of upper segment is zero. So, E_d will be equal to infinity at point L (Touching the y-axis).

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ii) At point A:-

$E_d = \frac{AD}{AL} > 1$. According to diagram, the length of

lower segment is greater than the length of upper segment at point A. So, the E_d will be greater than unity.

iii) At point B:-

$E_d = \frac{BD}{BL} = 1$. According to diagram, the length of

lower segment is same as length of upper segment at point B. So, the E_d will be equal to unity.

iv) At point C:-

$E_d = \frac{CD}{CL} < 1$. According to diagram, the length

of lower segment is less than the length of upper segment at point C. So, the E_d will be less than unity.



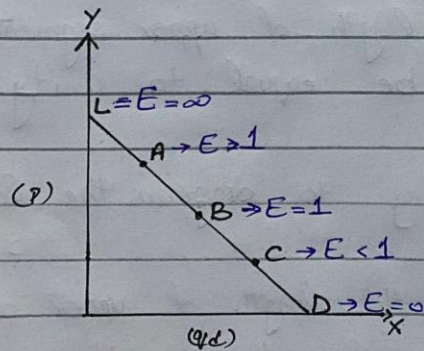
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v) At point D:-

$E_d = \frac{0}{DL} = 0$. According to diagram, the length

of the lower segment is equal to zero while the length of upper segment is DL. By putting these values in the formula we will get $E_d = 0$.

So,



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★ Other Concepts of the Elasticity of Demand

There are two other concepts of the elasticity of demand.

- i) Income Elasticity of Demand.
- ii) Cross Elasticity of Demand

i) Income Elasticity of Demand:-

It is the ratio of proportionate change in quantity demanded to the proportionate change in income

Formula:-

$$\Delta q = q_1 - q_0, \Delta Y = Y_1 - Y_0$$

$$E_x = \frac{\frac{\Delta q}{q}}{\frac{\Delta Y}{Y}}, \quad E_y = \frac{\Delta q}{\Delta Y} \times \frac{Y}{q}$$

$q_0 = 1^{\text{st}}$ Quantity
 $q_1 = 2^{\text{nd}}$ Quantity
 $Y_0 = 1^{\text{st}}$ Income
 $Y_1 = 2^{\text{nd}}$ Income

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* Main Points:-

- 1) E_y of normal goods is positive.
- 2) E_y of inferior goods is negative.
- 3) E_y can be greater, less or equal to unity (1).
- 4) With respect to luxuries $E_y > 1$.
- 5) With respect to necessities $E_y < 1$.
- 6) When ΔY does not affect q then, $E_y = 0$.
- 7) When ΔY is equal to Δq then, $E_y = 1$.

Eg.

Y	Qd	
50	10	$y_0 = 50$, $q_0 = 10$, $\Delta q = 30 - 10 = 20$
100	30	$y_1 = 100$, $q_1 = 30$, $\Delta Y = 100 - 50 = 50$

putting values in formula.

$$E_y = \frac{\Delta q}{\Delta Y} \times \frac{Y}{q} \rightarrow E_y = \frac{20}{50} \times \frac{50}{10} \rightarrow E_y = 2 > 1$$

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ii) Cross Elasticity of Demand:-

→ The rate of responsiveness of quantity demanded of product B to change in price of product A is known as cross elasticity of demand.

→ For example, wheat and Rice are substitutes. An increase in the price of Rice will affect demand for wheat and it will go up.

* Formula:-

$$E_c = \frac{\frac{\Delta q_B}{q_B}}{\frac{\Delta P_A}{P_A}}, \quad E_c = \frac{\Delta q_B}{\Delta P_A} \times \frac{P_A}{q_B}$$

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Factors Influencing the Elasticity of Demand

→ The following factors influence the elasticity of demand whether it is "more" or "less" elastic.

- (1) Nature of Good
- (2) Price Level
- (3) Different Uses of Goods
- (4) Substitute for the Good
- (5) Ability of the Good to Postpone its Use
- (6) Fashion, Habit & Attitude
- (7) Jointly Demanded Good
- (8) Relative Importance in Household Consumption
- (9) Level of Income

1) Nature of Good:-

The nature of different goods differ from each other. Thus their elasticity of demand also differs.



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- a) The elasticity of demand for necessities like Salt, Ghee is less elastic on the other hand luxuries are more elastic goods.
- b) Perishable goods like Fruits are less elastic while durable goods like Sofa Set are more elastic.

2) Price Level:-

Most precious goods like Diamonds and Giffen goods having a very low price like plastic shoes are less elastic or inelastic goods. Because a rise or fall in their prices do not affect demand of Richer or the poorer.



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3) Different Use of Goods:-

A good having only one specific use is less elastic. But goods having many uses have more elastic demand.

4) Substitute for the Good:-

Availability of close substitute of a good in the market, makes its demand more elastic. If the price of such a good rises, people start purchasing its substitute.

(5) Ability of the Good to Postpone it's Use:-

The goods whose demand can be postponed for some-time are known as more elastic goods.



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6) Fashion, Habit & Attitude:-

When some product "comes into Fashion" or "form a part of habit and taste" becomes less elastic or inelastic. People do not care about its price and prefer its purchase even at a higher price.

7) Jointly Demanded Goods:-

The goods which are jointly demanded are less elastic. As a rise or fall in their price may not influence their demand largely unless the price of complementary goods also change. Example, A fall in the price of petrol will have a very minor effect unless the price of cars also fall.



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